



## Selected Topics in Cryptography

### Lab Session 02: Toy ECDH

September 15, 2026

You can use C/C++, C#, Java or Python to develop your source code, but please use only one of this languages to develop all the exercises. You **MUST NOT USE any cryptographic library to implement ECDH**. Please use projective coordinates i.e  $(x, y, z)$  to represent each point of the elliptic curve.

## 1 Scalar multiplication $kP$

Develop the following exercises by your own. Please represent a point in an elliptic curve using three coordinates  $(x, y, z)$ . The point at infinity  $\mathcal{O} = (0, 1, 0)$ . Any other point in the elliptic curve will be represented as  $(x, y, 1)$ , where  $x, y \in \mathbb{Z}_p$ . Use this representation to do the following exercises.

**Use the implementation to add and double points in an elliptic curve that you did in lab session 1**

1. Design a function that implements **Algorithm 1** *Right-to-left binary method for point multiplication*

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**Algorithm 1:** Right-to-left binary method for point multiplication( $k, P$ )

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**Input** :  $k = (k_{t-1}, \dots, k_1, k_0)_2$  binary representation of  $k$ ;  $P \in \mathbb{E}(GF(p))$

**Output:**  $kP$

```
1  $Q = \infty$  ;                                     /* infinity point */
2 for  $i \leftarrow 0$  to  $t - 1$  do
3   if  $k_i == 1$  then
4      $Q \leftarrow Q + P$  ;                           /* point addition */
5   end
6    $P \leftarrow 2P$  ;                               /* point doubling */
7 end
8 return  $Q$ 
```

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2. Design a function that implements **Algorithm 2** *Left-to-Right binary method for point multiplication*
3. Test each function (exercise 1 and 2) for at least 5 different examples.

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**Algorithm 2:** Left-to-right binary method for point multiplication( $k,P$ )

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**Input :**  $k = (k_{t-1}, \dots, k_1, k_0)_2$  binary representation of  $k$ ;  $P \in \mathbb{E}(GF(p))$

**Output:**  $kP$

```
1  $Q = \infty$  ;                                /* infinity point */
2 for  $i \leftarrow t-1$  to 0 do
3    $Q \leftarrow 2Q$  ;                            /* point doubling */
4   if  $k_i == 1$  then
5      $Q \leftarrow Q + P$  ;                      /* point addition */
6   end
7 end
8 return  $Q$ 
```

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## 2 Simulation of ECDH

Simulate the ECDH with elliptic curves protocol in pairs. **Use the implementation to add and double points in an elliptic curve that you did in a previous lab session.** Also use the functions of Section 1 to compute  $kP$ . To show the simulation every student must use her/his own computer and proceed as follows.

1. Do agree with your partner a prime number  $p$  such that  $|p| \geq 7$  bits, the parameters of a non-singular elliptic curve  $a$  and  $b$  a generator point  $G \in \mathbb{E}(a, b)$ . Please consider that your program must accept large prime numbers, i.e.  $|p| = 128, 256, 512, 1024$  bits.
2. One student will play the role of Alice and the other one will play the role of Bob.
3. Alice randomly choose an integer  $k_A$  and compute  $A = k_A G$ . Share  $A$  with Bob.
4. Bob randomly choose exponent  $k_B$  and compute  $B = k_B G$ . Share  $B$  with Bob.
5. Once Alice received the value  $B$  from the other student, must calculate  $k_A B$
6. Once Bob received the value  $A$  from the other student, must calculate  $k_B A$
7. Compare your results.  $A$  and  $B$  must be equal.

## 3 Products

Every student can write her/his own report including the following information:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. The source code you did to implement algorithm 1 and algorithm 2 (Section 1)
3. Test your functions for Algorithm 1 and 2 as follows. Use at least 5 different examples. You must show the elliptic curve you used, the point  $P$ , the value of  $k$  and the output  $kP$ . Show the result of each iteration of the for cycle.